

AWS API Gateway Custom Domain: Part 1

To Begin with the Lab

Summary of the Lab

In this video, **AWS API Gateway Custom Domain** was introduced as a production-friendly alternative to the default long invoke URL. The lesson explained that a custom domain such as `https://api.company.com` improves branding, is easier to remember, and is better suited for real-world APIs. It also covered the main setup requirements, including owning a public domain, choosing whether the domain is public or private, selecting the API endpoint type, and attaching a valid **ACM** certificate. The video emphasized that public custom domains can be registered through **Route 53** or an external registrar, and that **mTLS** is available only with custom domains, not the default invoke URL. It also noted that routing mode is an important decision point and will be discussed separately.

Understanding API Gateway Custom Domains

A **custom domain** lets you replace the default API Gateway invoke URL with your own branded domain name. It exists so APIs look more professional, are easier for clients to remember, and can be configured more securely for production use. The setup depends on whether the domain is public or private, which API endpoint type you are using, and whether you want features like **mutual TLS authentication**. A valid **ACM** certificate is required, and the certificate region must match the endpoint type rules.

Example:

Instead of exposing an API with a long URL like `https://abc123.execute-api.us-east-1.amazonaws.com/prod`, a company uses `https://api.company.com`. The domain is pointed to **API Gateway** using an alias record in **Route 53** or a CNAME record in another DNS provider. If the company needs stronger security, it can also enable **mTLS** so only clients with a valid certificate can connect.

Lab Steps

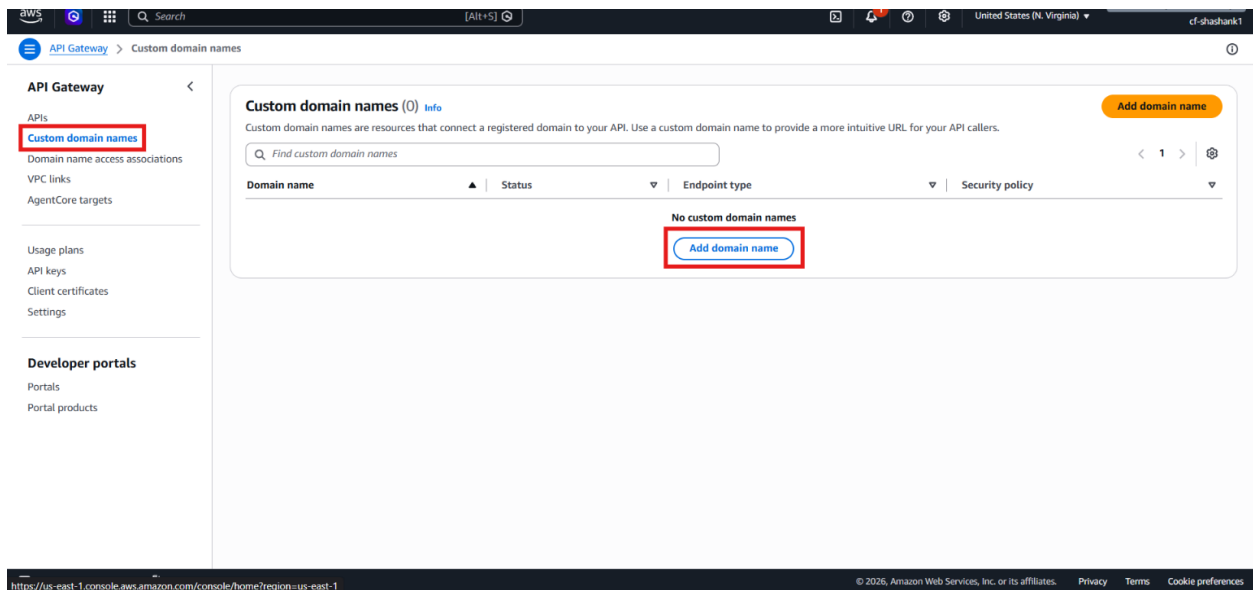
Step 1 – Understand the Domain Requirements

- Sign in to the AWS Management Console.
- Open **API Gateway**.
- Review the default invoke URL for the API.
- Understand that the custom domain will replace the long AWS-generated URL. `

Step 2 – Creating Custom Domain

- Open **API Gateway** → **Custom domain names**.

- Click on **Add domain name**.



- Make sure you own a public domain name such as api.company.com.

Step 3 – Prepare a Registered Domain Name

- Make sure you own a public domain name such as api.company.com.
- Use **Route 53** or an external registrar such as GoDaddy, Namecheap, or Cloudflare.
- If using **Route 53**, create an **Alias** record.
- If using an external registrar, create a **CNAME** record.

Step 4 – Choose the API Endpoint Type

- Select **Regional** if you are using a regional API.
- Select **Edge-Optimized** only if you want CloudFront-backed global distribution.
- Remember that **Regional** works with REST and HTTP APIs.
- Remember that **Edge-Optimized** is for REST APIs only.

Add domain name [Info](#)

Specify your domain details to create your domain name. Then, create a DNS record and map your APIs to your custom domain name.

Domain details

Domain name
Enter the subdomain or the root domain of your registered internet domain name.
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☒ **Public**
Create a domain that will be accessible from the public internet.
Supports REST, HTTP, and WebSocket APIs.

☐ **Private**
Create a private domain that's only accessible from within a VPC.
Supports only REST APIs.

Routing mode - new [Info](#)
Select the mode that API Gateway uses to send traffic to your custom domain name. The default mode is API mappings. Changing your routing mode doesn't modify existing routing rules or API mappings.

☒ **API mappings only**
Connect API stages to your custom domain name.
Supports REST, HTTP, and WebSocket APIs.

☐ **Routing rules only (recommended)**
Create rules on headers and base path conditions to connect API stages to your custom domain name.
Supports only REST APIs.

☐ **Routing rules then API mappings**
Use routing rules and API mappings to connect API stages to your custom domain name. In this mode all routing rules take priority over any API mappings.
Supports only REST APIs.

Public domain configuration

API endpoint type [Info](#)

☒ **Regional (recommended)**
Associate this custom domain name with a specific AWS Region to optimize intra-Region latency. Supports REST, HTTP, and WebSocket APIs.

☐ **Edge-optimized (only for REST APIs)**
Associate this custom domain name with an API endpoint that is replicated across AWS Regions using Amazon CloudFront. Supports only REST APIs.

Step 5 – Configure TLS and mTLS

- Choose the appropriate security policy.
- Use the policy that supports the required **TLS** versions, such as TLS 1.2 or TLS 1.3.
- Understand that **mTLS** is not available with the default invoke URL.
- Enable **mTLS** only if you need both server and client certificate validation on a custom domain.

Step 6 – Attach a Valid ACM Certificate

- Open **AWS Certificate Manager (ACM)**.
- Request or use a certificate that matches the custom domain name.
- Use a wildcard certificate if needed, such as *.company.com.
- For a **Regional** endpoint, make sure the certificate is in the same region as **API Gateway**.
- For an **Edge-Optimized** endpoint, make sure the certificate is in us-east-1.
- Create custom domain.

API Gateway > Custom domain names > Add domain name

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only REST APIs

IP address type [Info](#)
Select the type of IP addresses that can invoke your domain.

☒ **IPv4**
Supports REST, HTTP, and WebSocket APIs for public domains.

☐ **Dualstack**
Supports all domain types.

Mutual TLS authentication [Info](#)
Mutual TLS authentication requires two-way authentication between the client and the server. With mutual TLS, clients must present X.509 certificates to verify their identity to access your API. Supports REST and HTTP APIs.

☐ Use mutual TLS authentication

Security policy - new [Info](#)
Transport Layer Security (TLS) protects data in transit between a client and server. The security policy also determines the cipher suite options that clients can use with your API.

SecurityPolicy_TL13_1_2_2021_06

Endpoint access mode - new [Info](#)
Provide additional governance for your APIs.

☒ **Basic**
Allow all clients to access the domain.

☐ **Strict (recommended)**
Enforce Server Name Indication (SNI) validation.

ACM certificate [Info](#)
Select an AWS Certificate Manager certificate for your custom domain name. [Learn more](#)

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[Cancel](#) [Add domain name](#)

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Verification

- The purpose of **API Gateway Custom Domain** was understood successfully.
- The difference between the default invoke URL and a branded custom domain was explained.
- Public and private domain choices were identified.
- DNS setup requirements for **Route 53** and external registrars were noted.
- Endpoint type, **TLS**, **mTLS**, and **ACM** certificate rules were reviewed.
- The lesson prepared the foundation for custom-domain routing mode in the next video.